

Adios, BIOS!

The familiar beep you hear when you turn on your computer may soon be gone. Here's its new incarnation

The Basic Input Output System (BIOS) has been around for quite some time, and you have probably seen it on every PC you have powered on. Press [Del] or [F1], and there it is—the blue screen with all those exciting tweaking options. You probably know what the BIOS does—it makes your computer evil.

No, seriously, the BIOS does things such as checking the hardware, checking if the RAM is functional, and so on. But as just another piece of software, it's seriously limited in features, according to industry experts. And the same experts also say we've lived with it long enough, and it's time to move on.

Enter Microsoft and Intel

Extensible Firmware Interface, or EFI, is the new framework that Intel and Microsoft have worked on. It will perform all the functions that the BIOS does, and pitch in some more. The two companies will start an industry forum this year to officially murder the BIOS as we know it, and cause the impending birth of EFI. The first PC with the EFI interface—a Gateway PC—has already shipped, in 2003.

Essentially, the EFI specification is a pre-boot environment, which allows a computer to run diagnostics, scan for viruses, etc.

If the BIOS is an interface between the OS and the platform firmware and hardware, the EFI specification defines a new model for this: It comprises data tables holding platform-related information, as well as boot and runtime service calls available to the OS and its loader. These provide a new, standard environment for booting up an OS and running pre-boot applications.

Intel believes that promoting the specification as a standard will help PC manufacturers and please PC users—by making computers start up more quickly; by improving the ability to manage PCs and servers remotely; and by helping hardware makers cut manufacturing and support costs. For example, EFI PCs will be able to run diagnostic utilities before loading their OSes.

Why EFI?

To quote Mike Richmond, of Intel's Software and Solutions Group: "In the long run, we can bring boot time down, because the EFI framework

